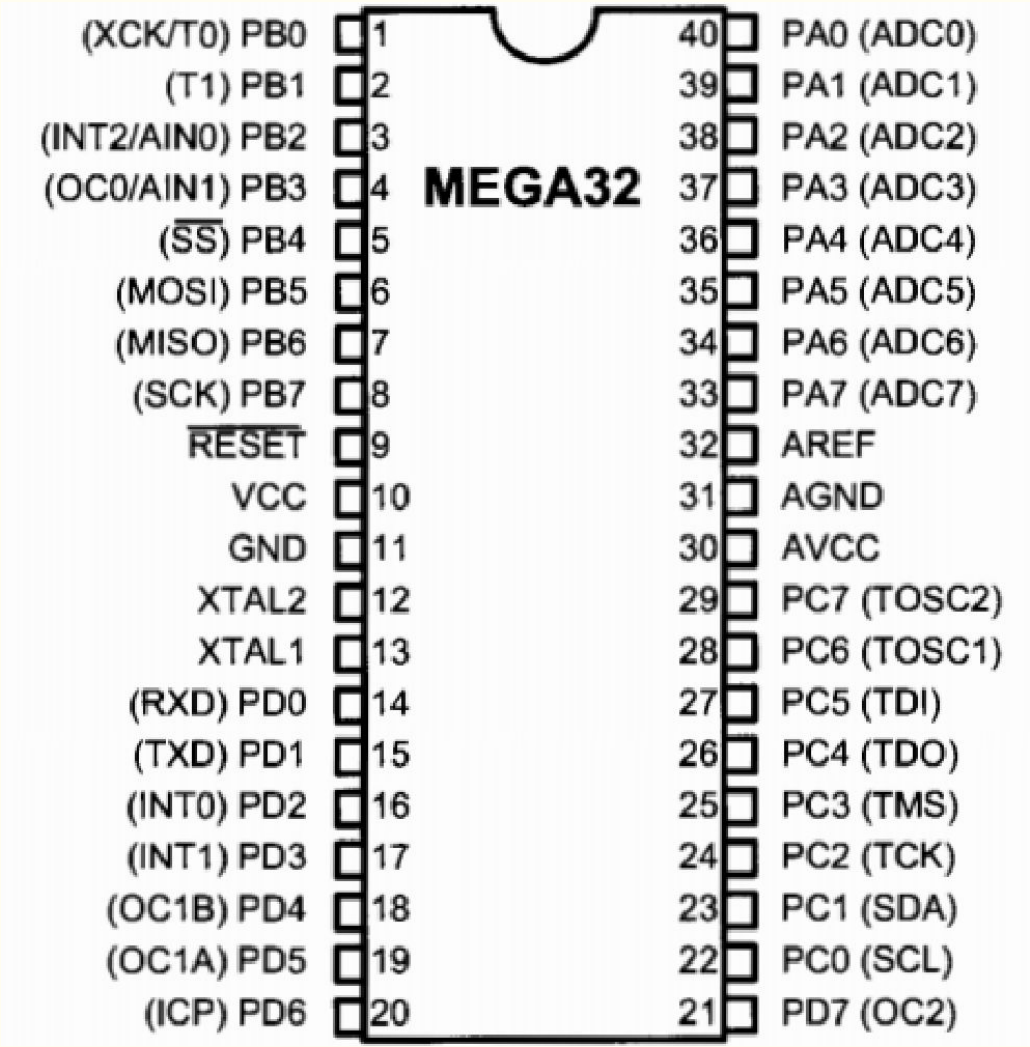
ATmega32 Pinout and Timer I/O Registers

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Chapter 4-Fig4-1



Bit	7	6	5	4	3	2	1	0
	FOC0	WGM00	COM01	COM00	WGM01	CS02	CS01	CS00
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC0 D7 Force Output compare: Writing 1 to it forces the wave generator to act as if a compare match has occurred.

WGM01:00	D6	D3	Timer0 mode selector bits
	0	0	Normal
	0	1	CTC (Clear Timer on Compare match)
	1	0	PWM, phase correct
	1	1	Fast PWM

COM01:00 D5 D4 Compare Output Mode; The table shows what the wave generator does on compare match when the timer is in Normal or CTC mode:

COM01	COM00	Description
0	0	Normal port operation, OC0 disconnected
0	1	Toggle OC0 on compare match
1	0	Clear OC0 on compare match
1	1	Set OC0 on compare match

CS02:00	D2	D1	D0	Timer0 clock selector
	0	0	0	No clock source (Timer/Counter stopped)
	0	0	1	clk (no prescaling)
	0	1	0	clk / 8
	0	1	1	clk / 64
	1	0	0	clk / 256
	1	0	1	clk / 1024
	1	1	0	External clock source on T0 pin. Clock on falling edge
	1	1	1	External clock source on T0 pin. Clock on rising edge

Figure 15-4. TCCR0 (Timer/Counter Control Register) Register

D7							D0
OCIE2	TOIE2	TICIE1	OCIE1A	OCIE1B	TOIE1	OCIE0	TOIE0

TOIE0 Timer0 overflow interrupt enable
= 0 Disables Timer0 overflow interrupt
= 1 Enables Timer0 overflow interrupt

OCIE0 Timer0 output compare match interrupt enable
= 0 Disables Timer0 compare match interrupt
= 1 Enables Timer0 compare match interrupt

TOIE1 Timer1 overflow interrupt enable
= 0 Disables Timer1 overflow interrupt
= 1 Enables Timer1 overflow interrupt

OCIE1B Timer1 output compare B match interrupt enable
= 0 Disables Timer1 compare B match interrupt
= 1 Enables Timer1 compare B match interrupt

OCIE1A Timer1 output compare A match interrupt enable
= 0 Disables Timer1 compare A match interrupt
= 1 Enables Timer1 compare A match interrupt

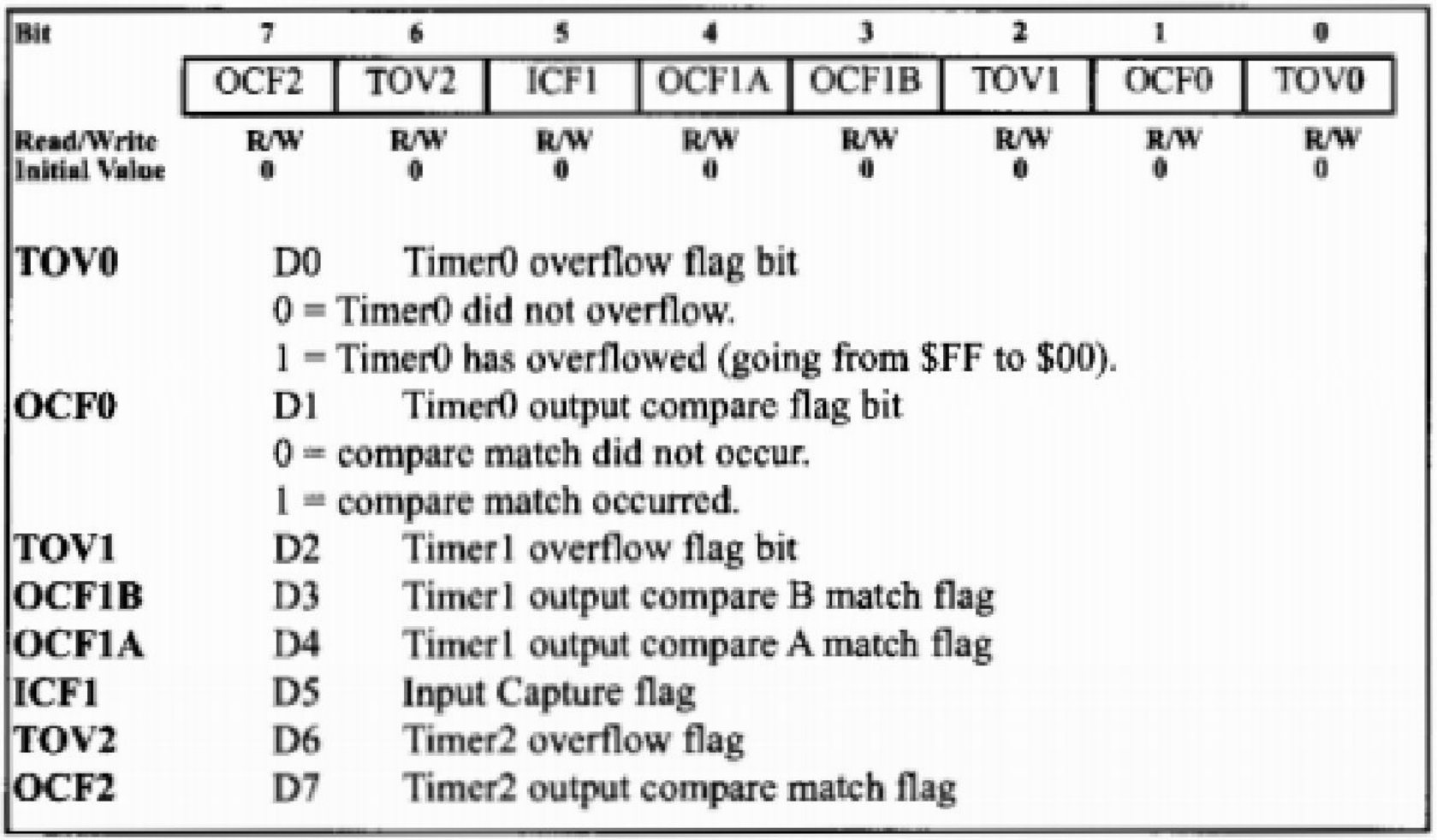
TICIE1 Timer1 input capture interrupt enable
= 0 Disables Timer1 input capture interrupt
= 1 Enables Timer1 input capture interrupt

TOIE2 Timer2 overflow interrupt enable
= 0 Disables Timer2 overflow interrupt
= 1 Enables Timer2 overflow interrupt

OCIE2 Timer2 output compare match interrupt enable
= 0 Disables Timer2 compare match interrupt
= 1 Enables Timer2 compare match interrupt

These bits, along with the I bit, must be set high for an interrupt to be responded to. Upon activation of the interrupt, the I bit is cleared by the AVR itself to make sure another interrupt cannot interrupt the microcontroller while it is servicing the current one. At the end of the ISR, the RETI instruction will make I = 1 to allow another interrupt to come in.

Figure 10-3. TIMSK (Timer Interrupt Mask) Register



Bit	7	6	5	4	3	2	1	0
	OCF2	TOV2	ICF1	OCF1A	OCF1B	TOV1	OCF0	TOV0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

TOV0 D0 Timer0 overflow flag bit
0 = Timer0 did not overflow.
1 = Timer0 has overflowed (going from $FF to $00).

OCF0 D1 Timer0 output compare flag bit
0 = compare match did not occur.
1 = compare match occurred.

TOV1 D2 Timer1 overflow flag bit

OCF1B D3 Timer1 output compare B match flag

OCF1A D4 Timer1 output compare A match flag

ICF1 D5 Input Capture flag

TOV2 D6 Timer2 overflow flag

OCF2 D7 Timer2 output compare match flag

Figure 9-6. TIFR (Timer/Counter Interrupt Flag Register)

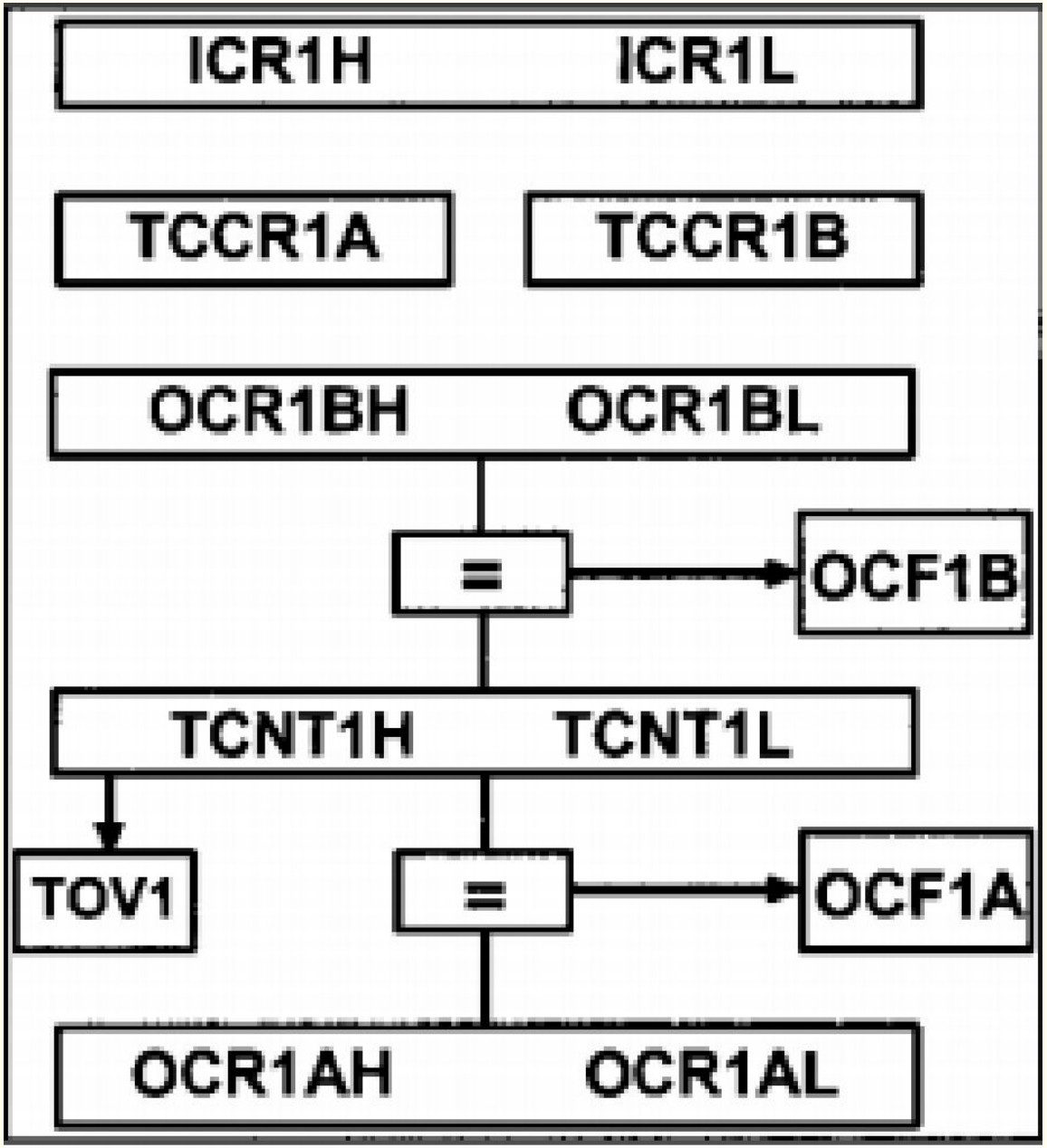
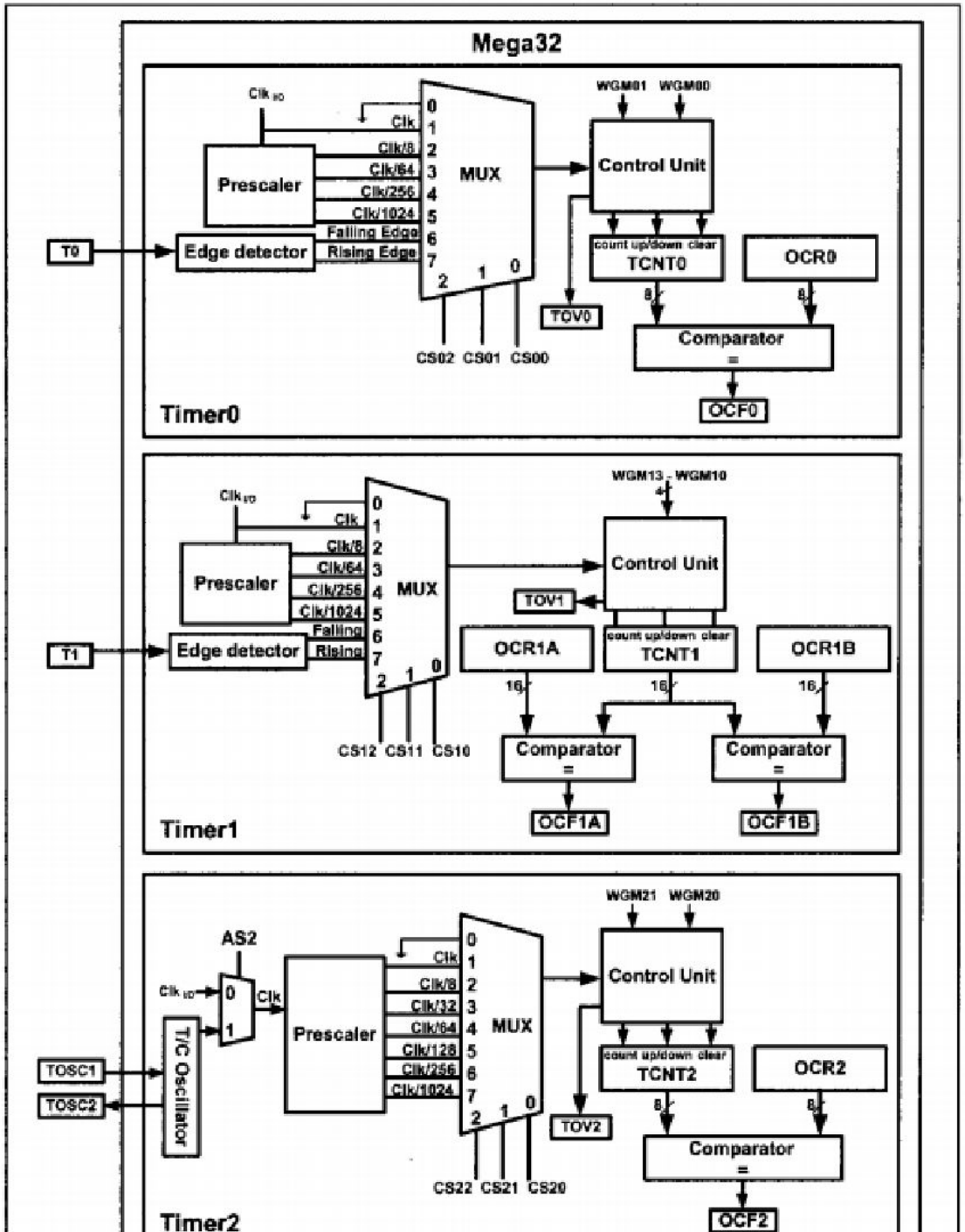
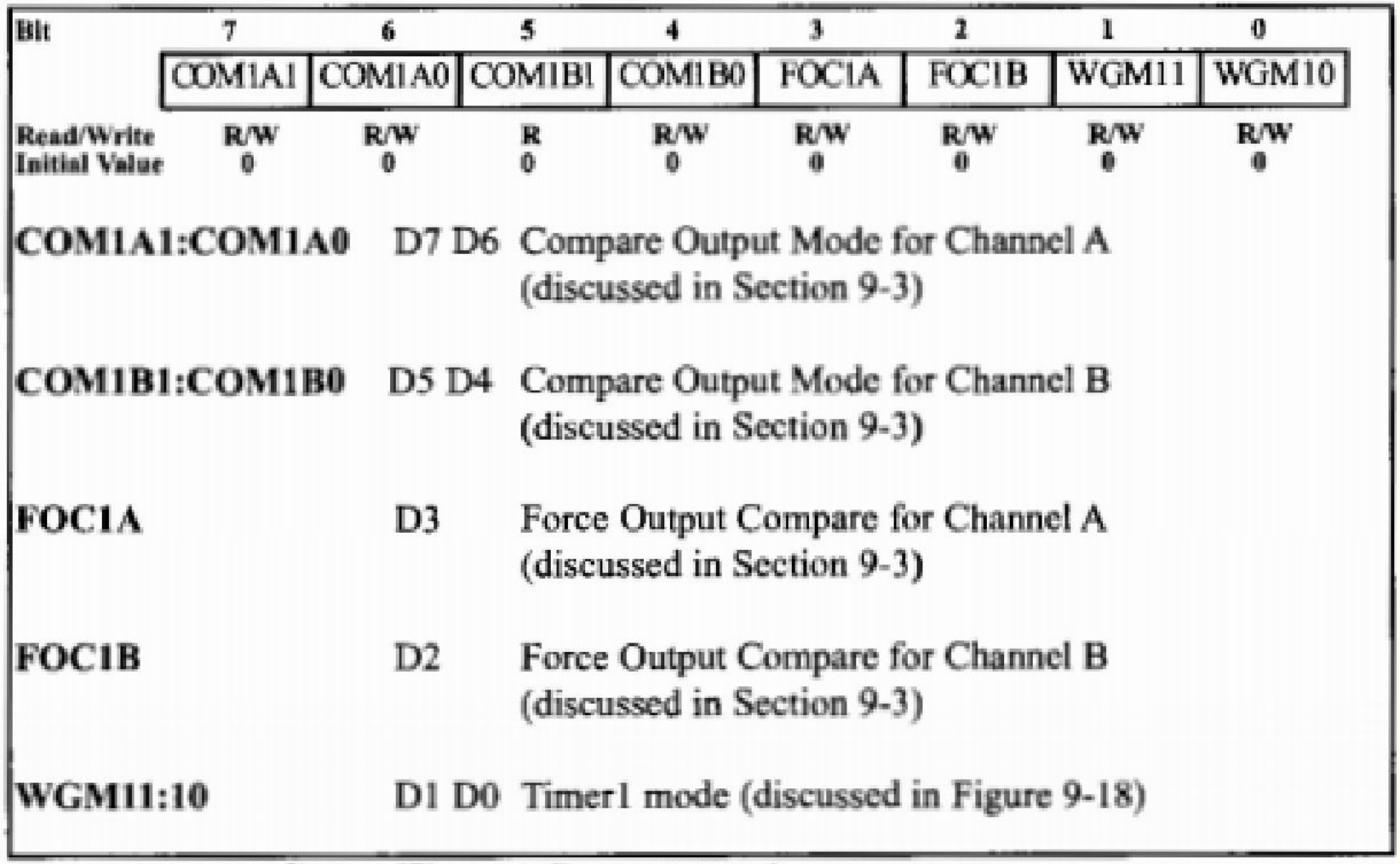


Figure 9-14. Simplified Diagram of Timer1


Mega32
Clk I/O
Clk
Clk/8
Clk/64
Clk/256
Clk/1024
Falling Edge
Rising Edge
Prescaler
Edge detector
T0
MUX
WGM01 WGM00
Control Unit
count up/down clear
TCNT0
OCR0
TOV0
Comparator
=
OCF0
CS02 CS01 CS00
Timer0
WGM13 - WGM10
Clk I/O
Clk
Clk/8
Clk/64
Clk/256
Clk/1024
Falling
Rising
Prescaler
Edge detector
T1
MUX
Control Unit
TOV1
OCR1A
count up/down clear
TCNT1
OCR1B
16
Comparator
=
OCF1A
OCF1B
CS12 CS11 CS10
Timer1
WGM21 WGM20
AS2
Clk I/O
Clk
Clk/8
Clk/32
Clk/64
Clk/128
Clk/256
Clk/1024
Prescaler
MUX
Control Unit
TOSC1
TOSC2
T/C Oscillator
count up/down clear
TCNT2
OCR2
TOV2
Comparator
=
OCF2
CS22 CS21 CS20
Timer2



Bit	7	6	5	4	3	2	1	0
	COM1A1	COM1A0	COM1B1	COM1B0	FOC1A	FOC1B	WGM11	WGM10
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

COM1A1:COM1A0 D7 D6 Compare Output Mode for Channel A (discussed in Section 9-3)

COM1B1:COM1B0 D5 D4 Compare Output Mode for Channel B (discussed in Section 9-3)

FOC1A D3 Force Output Compare for Channel A (discussed in Section 9-3)

FOC1B D2 Force Output Compare for Channel B (discussed in Section 9-3)

WGM11:10 D1 D0 Timer1 mode (discussed in Figure 9-18)

Figure 9-17. TCCR1A (Timer 1 Control) Register

Bit	7	6	5	4	3	2	1	0	
	ICNC1	ICES1	-	WGM13	WGM12	CS12	CS11	CS10	TCCR1B
Read/Write	R/W	R/W	R	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

ICNC1 D7 Input Capture Noise Canceler
0 = Input Capture is disabled.
1 = Input Capture is enabled.

ICES1 D6 Input Capture Edge Select
0 = Capture on the falling (negative) edge
1 = Capture on the rising (positive) edge

D5 Not used

WGM13:WGM12 D4 D3 Timer1 mode

Mode	WGM13	WGM12	WGM11	WGM10	Timer/Counter Mode of Operation	Top	Update of OCR1x	TOV1 Flag Set on
0	0	0	0	0	Normal	0xFFFF	Immediate	MAX
1	0	0	0	1	PWM, Phase Correct, 8-bit	0x00FF	TOP	BOTTOM
2	0	0	1	0	PWM, Phase Correct, 9-bit	0x01FF	TOP	BOTTOM
3	0	0	1	1	PWM, Phase Correct, 10-bit	0x03FF	TOP	BOTTOM
4	0	1	0	0	CTC	OCR1A	Immediate	MAX
5	0	1	0	1	Fast PWM, 8-bit	0x00FF	TOP	TOP
6	0	1	1	0	Fast PWM, 9-bit	0x01FF	TOP	TOP
7	0	1	1	1	Fast PWM, 10-bit	0x03FF	TOP	TOP
8	1	0	0	0	PWM, Phase and Frequency Correct	ICR1	BOTTOM	BOTTOM
9	1	0	0	1	PWM, Phase and Frequency Correct	OCR1A	BOTTOM	BOTTOM
10	1	0	1	0	PWM, Phase Correct	ICR1	TOP	BOTTOM
11	1	0	1	1	PWM, Phase Correct	OCR1A	TOP	BOTTOM
12	1	1	0	0	CTC	ICR1	Immediate	MAX
13	1	1	0	1	Reserved	-	-	-
14	1	1	1	0	Fast PWM	ICR1	TOP	TOP
15	1	1	1	1	Fast PWM	OCR1A	TOP	TOP

CS12:CS10 D2D1D0 Timer1 clock selector

0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (no prescaling)
0	1	0	clk / 8
0	1	1	clk / 64
1	0	0	clk / 256
1	0	1	clk / 1024
1	1	0	External clock source on T1 pin. Clock on falling edge.
1	1	1	External clock source on T1 pin. Clock on rising edge.

Figure 9-18. TCCR1B (Timer 1 Control) Register

Bit	7	6	5	4	3	2	1	0
	FOC2	WGM20	COM21	COM20	WGM21	CS22	CS21	CS20
Read/Write	W	RW	RW	RW	RW	RW	RW	RW
Initial Value	0	0	0	0	0	0	0	0

FOC2 D7 Force compare match: a write-only bit, which can be used while generating a wave. Writing 1 to it causes the wave generator to act as if a compare match had occurred.

WGM20, WGM21

D6	D3	Timer2 mode selector bits
0	0	Normal
0	1	CTC (Clear Timer on Compare Match)
1	0	PWM, phase correct
1	1	Fast PWM

COM21:20 D5 D4 Compare Output Mode:
These bits control the waveform generator (see Chapter 15).

CS22:20

D2	D1	D0	Timer2 clock selector
0	0	0	No clock source (Timer/Counter stopped)
0	0	1	clk (No Prescaling)
0	1	0	clk / 8
0	1	1	clk / 32
1	0	0	clk / 64
1	0	1	clk / 128
1	1	0	clk / 256
1	1	1	clk / 1024

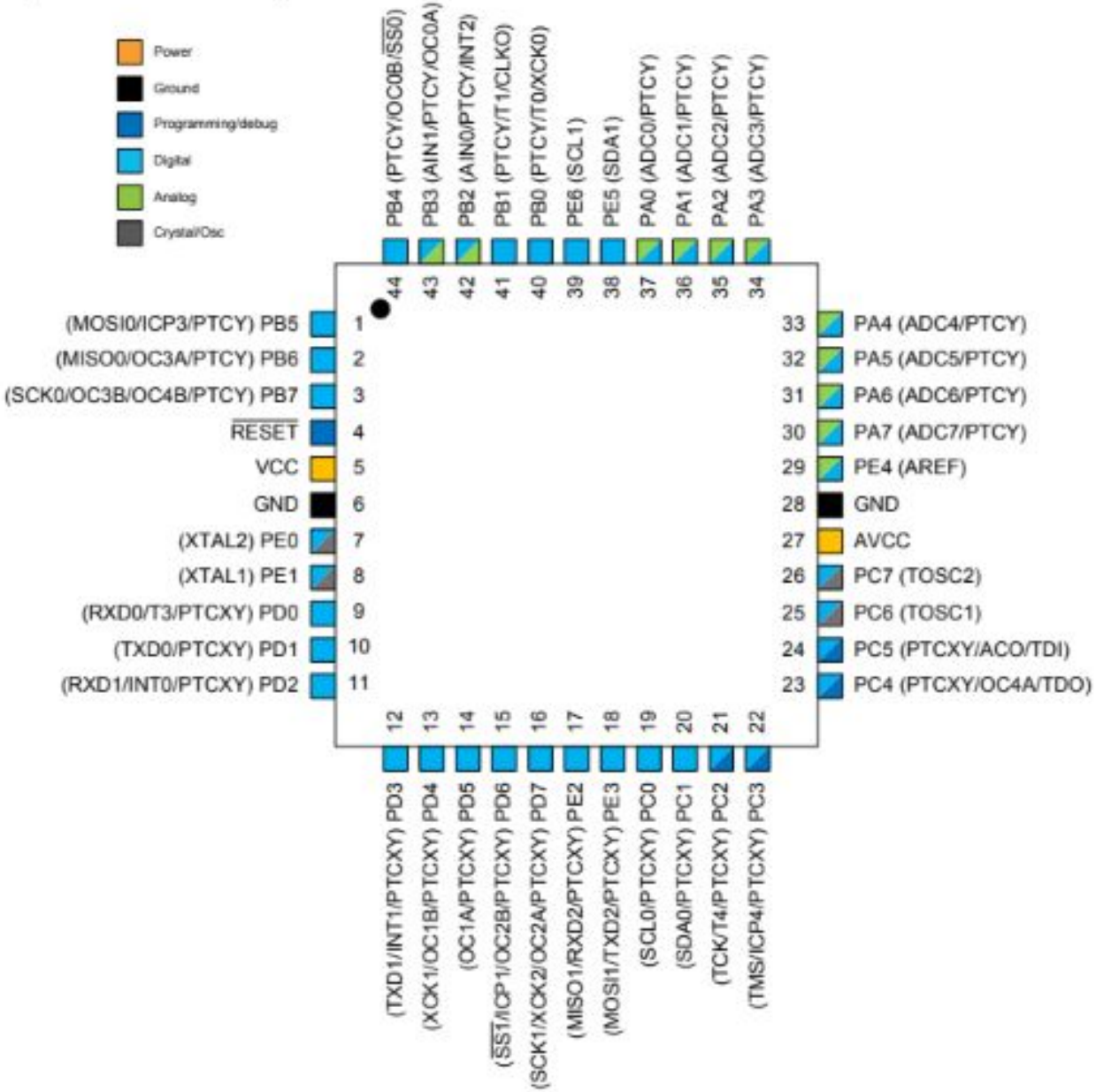
Figure 9-11. TCCR2 (Timer/Counter Control Register) Register

ATmega324pb Pinout and Timer I/O Registers

- ATmega324pb Datasheet
- Timers 0, 1, 2, 3, 4
-

Pin Configurations

Figure 5-1. Pinout ATmega324PB



ATmega324pb Pinout and Timer I/O Registers

- ATmega324pb Datasheet
- Timer 0
  - TCCR0A (pp. 126-129)
  - TCCR0B (pp. 129-130) (contains CS02:00 for normal timer operation/prescaling and for timer/event counter)
  - TCNT0 (same as for ATmega32)
  - TIFR0 (like TIFR but only for Timer 0)